

AI for Legal Assistance & Access

Making legal documents readable, comparable, multilingual, and safer to act on.

PRODUCT REQUIREMENTS DOCUMENT

Version 2.1 | Compact implementation revision | September 16, 2026

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Core principle: the LLM is the reasoning layer, not the source of truth. The document, retrieved legal sources, structured metadata, and citation verifier form the evidence layer.

Executive Summary

NyayaSetu addresses a practical problem identified in the supplied PRD: people often receive contracts, rental agreements, offer letters, loan terms, and insurance documents that they cannot comfortably understand. The product combines document parsing, RAG, structured risk analysis, multilingual support, comparison, and action-oriented outputs while keeping a clear boundary between legal information and professional legal advice.

What the user can do

Understand	Check	Ask	Act
TL;DR, section summaries, clause explanations	Risk tags, missing terms, key dates	Grounded Q&A; with exact source clauses	Checklist, questions, timeline, lawyer brief

Part	Coverage
1	Problem, scope, users and goals
2	Feature and workflow specification
3	Concrete AI model assignment
4	UI, architecture and RAG pipeline
5	Multilingual, API and data contracts
6	Tech stack, repository and evaluation
7	Guardrails, risks, roadmap and team

Layout correction: the original 18-page file contained several pages with a small table or diagram followed by a large unused region. This revision consolidates related sections, keeps diagrams close to their explanation, and uses the available page area for implementation details.

1. Problem, Scope, Goals and Users

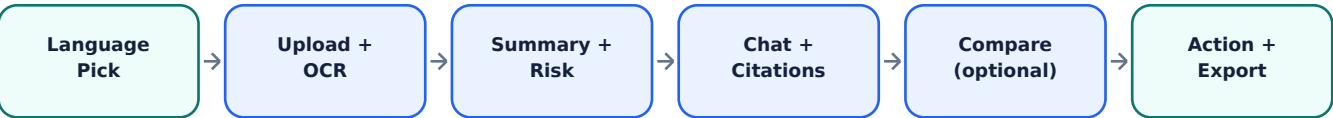
The product problem is the combination of language gap, access gap and trust gap described in the original PRD. The opportunity is not to replace lawyers, but to help a user understand a document, identify questions, and recognize when professional help is appropriate.

Goals	Non-goals / boundaries
Plain-language understanding in seconds; obligations and deadlines; risk flags; document-grounded Q&A; comparison; Indian regional languages; actionable checklists.	No binding legal drafting; no case-outcome prediction; no definitive legal advice; v1 scoped to Indian rental, employment and consumer agreement contexts; no default long-term vault.

Target users

User	Need	Primary workflow
Riya, first-time renter	Understand deposit, lock-in and termination clauses.	Upload -> risk -> Q&A;
Aman, early-career engineer	Compare notice period, non-compete and bonus terms.	Two-doc compare
Sunita, freelancer / small business owner	Review vendor contracts, payment and liability.	Risk -> action center
Mahesh, non-English-first consumer	Understand loan or insurance documents in Marathi.	Multilingual -> Q&A;

Primary journey



2. Feature Specification and Product Logic

Feature	Product behavior	AI / deterministic logic
Simplify	Three depths: TL;DR, section-wise, full annotated.	Gemini 3.8 Flash with structured output; preserve clause IDs.
Risk & Clause Highlighter	Standard, Unusual, High-Risk, Missing-but-expected.	Rules + LLM classification. Rules handle measurable patterns; LLM explains context.
Document Q&A;	Answers only from uploaded document unless explicitly routed to Legal Literacy.	Hybrid retrieval -> rerank -> LLM -> citation verifier.
Compare Mode	Show clause differences and evidence. Avoid an opaque single “winner”.	Clause alignment + structured diff + user-selected comparison criteria.
Action Center	Checklist, deadlines, red flags, lawyer questions and export.	Structured extraction + deterministic date/timeline formatting.
Multilingual Layer	User can ask and receive output in a selected language.	IndicLID + IndicTrans2 + multilingual LLM generation; verify against original source.
Legal Literacy	General legal concepts and rights questions, separated from private-document chat.	Indian legal-source RAG + jurisdiction filter + LLM.

Key product correction

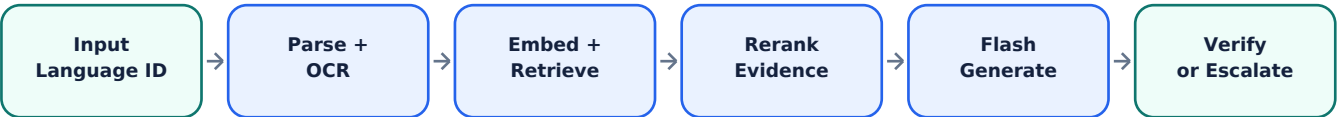
The original Compare Mode used the phrase “verdict: Offer B is more favorable.” For implementation, replace that with an evidence-led comparison. Example: “Offer A has a 90-day notice period; Offer B has 60 days. Offer B has a shorter stated non-compete and a guaranteed bonus floor.” If the user wants a decision framework, let the user choose the criteria and show the trade-offs rather than hiding them behind one score.

This makes the comparison auditable and consistent with the product's legal-comprehension purpose.

3. AI Model Map: Exactly Where Each Model Goes

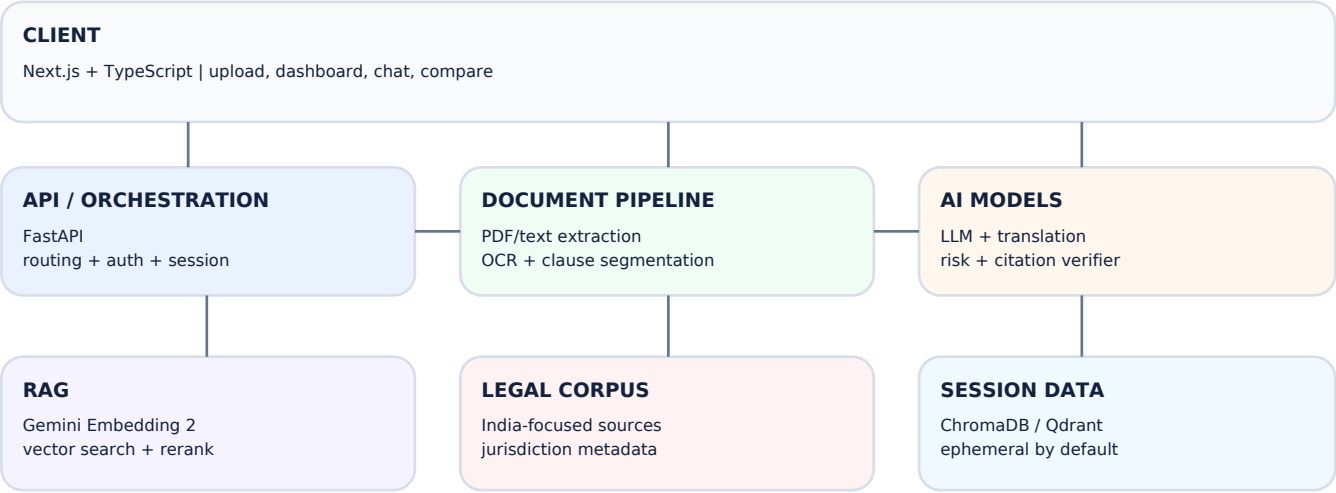
Stage	Model / method	Use it for	Do not use it for
OCR / layout	Google Document AI; native PDF extraction first	Scanned docs, tables, page layout, source spans	Legal reasoning
Language detection	AI4Bharat IndicLID	Source language and user/chat language	Translation or answer generation
Main reasoning	Gemini 3.8 Flash	Summaries, Q&A;, action extraction, normal compare/risk pass	Replacing retrieval or evidence
Hard cases	Gemini 3.1 Pro Preview	Ambiguous/high-impact cases that need a deeper second pass	Every request, due to cost/latency
Embeddings	Gemini Embedding 2	Clause indexing and multilingual semantic retrieval	Generating answers
Reranking	BAAI bge-reranker-v2-m3	Reordering retrieved clauses before generation	Final legal conclusion
Translation	IndicTrans2	Explicit English <-> Indic and Indic <-> Indic translation	Source of legal truth
Risk classifier	Rules + Gemini 3.8 Flash JSON schema	Risk label + reason + evidence	Unverified free-form prose
Citation verifier	Gemini 3.8 Flash verification pass	Check that cited source actually supports claim	Adding new facts

Recommended routing



Routing rule: keep the normal path on Flash. Escalate only when evidence is weak, the clause is ambiguous, or the comparison requires deeper reasoning. Never use a larger model to compensate for poor retrieval.

4. System Architecture and UI



All user-facing claims pass through scope + citation verification before rendering.

Screen	Key UI elements	Data shown
Summary Dashboard	TL;DR, risk summary, dates, clause map	Structured summary + risk JSON + date fields
Clause Detail + Chat	Original clause, explanation, chat, citation cards	Original source span + retrieved evidence + answer
Compare	Aligned clauses, changed values, criteria chips	Two parsed documents + diff objects
Action Center	Timeline, checklist, questions, export	Dates, obligations, risk reasons

Compact dashboard example

DOCUMENT	RISK	KEY DATES
Rental Agreement 11-month lease INR 15,000/month 2-month deposit	MEDIUM 3 unusual 1 high-risk 1 missing	Move-in: 1 Oct 2026 Renewal: 30 days prior Refund: 15 days

The original wireframe page used three small mockups with a large unused lower half. In the revised layout, the dashboard, clause/chat behavior, and comparison logic are represented together so the page communicates both design and implementation.

5. RAG, Citation and Risk Pipeline



Step	Implementation	Failure behavior
1. Extract	Create doc_id, page, clause_id, source span, original text.	If OCR confidence is low, mark section for review.
2. Segment	Split by headings, clause boundaries and semantic units, not arbitrary fixed chunks alone.	Keep original page/span metadata.
3. Retrieve	Semantic embeddings + lexical search for exact legal terms.	Return “insufficient evidence” if no relevant span passes threshold.
4. Rerank	bge-reranker-v2-m3 over top candidates.	Drop weak evidence before generation.
5. Generate	LLM receives question + selected evidence + output language.	No evidence means no document-specific claim.
6. Verify	Check every citation against the cited source span.	Block, revise or downgrade confidence when unsupported.

Risk classifier contract

Risk JSON	Example	Meaning
clause_id	3	Stable UI and citation reference
risk_level	high	standard / unusual / high / missing
reason	Full deposit forfeiture	Plain-language rationale
source	page 4, span 412-588	Auditable evidence location
confidence	high	Model confidence, not legal certainty

6. Multilingual Architecture and Legal Literacy

Multilingual support is a pipeline, not a single translation call. The original-language clause remains canonical. The user-facing explanation can be localized, but citations always resolve to the original evidence.

Stage	Component	Rule
Detect	IndicLID	Detect native-script and romanized Indic input.
Parse	OCR / native extraction	Create source spans before any translation.
Retrieve	Gemini Embedding 2	Retrieve across languages from the same semantic space.
Generate	Gemini 3.8 Flash	Answer in the user's selected language.
Translate	IndicTrans2	Use for explicit translation and localized copies.
Verify	Original-source verification	Verify meaning against canonical source text.

Example flow

Marathi question -> IndicLID identifies Marathi -> retrieve an English clause -> rerank evidence -> Gemini answers in Marathi -> citation card points to the original English clause -> optional translated clause appears separately.

Language rollout

- Hackathon UI: English, Hindi, Marathi, Tamil, Bengali, Gujarati, Telugu, Kannada and Malayalam.
- Architecture expansion: IndicTrans2 supports all 22 scheduled Indic languages, so additional languages can be enabled without changing the core routing design.
- Quality requirement: evaluate legal terminology and meaning preservation per language. Do not assume equal quality across languages.

Legal Literacy routing

Questions such as “What are my rights as a tenant in Maharashtra?” should not use the uploaded-document index. Route them to a separate Indian legal-source corpus with jurisdiction metadata, then generate a sourced informational answer with the same disclaimer.

7. API Contracts and Data Models

Endpoint	Method	Purpose	AI path
/upload	POST	PDF/DOCX/image -> parsed document	OCR + language ID + segmentation
/summarize/{docId}	GET	TL;DR + section + clause summaries	Gemini Flash
/risk/{docId}	GET	Structured clause risk	Rules + Gemini Flash
/qa/{docId}	POST	Grounded answer + citations	Embed + rerank + Gemini + verify
/compare	POST	Two-document diff	Clause alignment + Gemini
/action-center/{docId}	GET	Checklist, dates, lawyer brief	Structured extraction
/translate	POST	Explicit output translation	IndicTrans2 / controlled LLM

Canonical document object

```
{"doc_id":"uuid","source_language":"eng_Latn","jurisdiction":"IN","clauses":[{"clause_id":"3","page":4,"text":"...", "span":[412,588],"risk":"high"}]}
```

Grounded Q&A; response

```
{"answer":"...", "language":"mar", "confidence":"high", "citations":[{"clause_id":"3", "page":4, "span":[412,588]}], "verified":true}
```

Session rules

- Every request carries doc_id and jurisdiction metadata.
- Every citation carries page + clause_id + source span.
- No persistent document storage unless the user opts in.
- Translated text never overwrites canonical source text.

8. Tech Stack and Repository

Layer	Choice	Role
Frontend	Next.js + TypeScript + Tailwind + shadcn/ui	Dashboard, clause cards, chat, compare, export
Backend	FastAPI + Pydantic	Async API, validation, model routing
PDF / OCR	PyMuPDF + Google Document AI fallback	Digital extraction + scanned layout OCR
LLM	Gemini 3.8 Flash + selective Gemini 3.1 Pro Preview	Normal generation + escalation path
Embeddings	Gemini Embedding 2	Multilingual semantic retrieval
Reranker	bge-reranker-v2-m3	Evidence ranking
Translation	IndicTrans2	22 Indic languages
Vector store	ChromaDB for demo; Qdrant/Pinecone later	Session isolation and scalable retrieval
Auth/storage	Supabase, opt-in persistence	Auth + Postgres when storage is enabled

Repository structure

Folder	Key files
frontend/	app/, components/, lib/api.ts
backend/api/	upload.py, summarize.py, risk.py, qa.py, compare.py, action.py, translate.py
backend/pipeline/	ocr.py, parser.py, clauses.py, embedder.py, retriever.py, reranker.py, citation_verifier.py
backend/services/	llm.py, translation.py, legal_corpus.py, language_router.py
backend/guardrails/	disclaimer.py, scope_filter.py, policy_checks.py
eval/	citation/, risk/, multilingual/
data/	sample_documents/, legal_sources/ metadata

9. Evaluation, Metrics and Guardrails

Metric	Target	Test method
Comprehension lift	+2 points	Pre/post user rating
Time to clarity	<30 sec	Upload to first useful summary/risk view
Citation accuracy	>95%	Evidence entailment audit
Risk precision	>85% agreement	Legal-reviewer audit
Groundedness	>95% supported claims	Claim-to-source mapping
Demo completion	<3 min	Upload -> summary -> Q&A; -> export
Multilingual quality	Per-language tracking	Human legal terminology review

Responsible AI guardrails

Risk	Mitigation	Gate
Hallucination	RAG + reranking + citation verification	Unsupported answers blocked
Advice overreach	Persistent disclaimer and careful wording	No definitive sign/sue/win instructions
Jurisdiction mismatch	Jurisdiction metadata + source filter	Out-of-scope query flagged
Data leakage	Ephemeral sessions + opt-in persistence	No default long-term storage
OCR errors	Confidence score + review flag	Low-confidence text cannot silently become high-confidence output
Translation drift	Canonical source + translation + verification	Citation always resolves to original text

Session disclaimer

"This is legal information, not legal advice. Consult a licensed advocate for decisions with legal consequence."

10. Roadmap, Team and Implementation Checklist

Day	Build	Done when
Day 1	OCR/parse -> clause segmentation -> summary -> risk	Rental document works end-to-end
Day 2	RAG chat -> citations -> multilingual -> compare	Marathi/English grounded Q&A; and two-doc comparison work
Day 3	Action Center -> export -> evaluation -> UI polish	Live demo completes under 3 minutes

Role	Focus
AI/ML Lead	RAG, model routing, prompts, risk schema, evaluation
Backend	FastAPI, OCR/parser, vector store, retrieval, verifier
Frontend	Next.js, clause cards, risk badges, chat, compare, export
Product / Design	User flow, sample documents, legal-source curation, demo and disclaimer

Hackathon implementation checklist

Priority	Task	Acceptance check
P0	Upload + parse	Every clause has page and source span
P0	RAG Q&A;	Answer cannot appear without evidence
P0	Risk JSON	UI renders deterministic badges from schema
P0	Citation verifier	Unsupported citation blocks answer
P1	Marathi + English	Question, answer and UI labels work in both
P1	Compare	Differences are traceable to both documents
P1	Action Center	Dates/checklist/export generated from structured fields
P2	Legal Literacy	Uses separate legal-source RAG

Final implementation principle

Build NyayaSetu as an evidence-first legal comprehension system. Use specialized models for specialized jobs, keep the original clause canonical, retrieve before generating, verify citations before rendering, and treat multilingual output as a presentation layer rather than a replacement for source evidence.

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